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MIL-A-50687A (AR)
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SUPERSEDING
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15 April 1971

MILITARY SPECIFICATION

ANTENNA ASSEMBLY AS-2049/VPS-2: 10556312

This specification is approved for use within the US Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers Antenna Assembly AS-2049/VPS-2 which is used as the antenna for the Radar Set AN/VPS-2 (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standars and handbooks. The following specifications standards and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-F-13926	-	Fire Control, Material, General Specification Governing the Manufacture and Inspection of
MIL-I-45607	-	Inspection Equipment, Acquisition, Maintenance and Disposition of
MIL-S-45743	-	Soldering, Manual Type, High Reliability, Electrical and Electronic Equipment

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document, should be addressed to: Commander U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-S-46844	-	Soldering, Bath Soldering of Printed Wiring Assemblies
MIL-R-48427	-	Reflector and Feed Assembly, 10556313
MIL-S-50666	-	Servo Drive, Traverse, 10548052
MIL-S-50667	-	Servo Drive, Elevation, 10548062

STANDARDS

MILITARY

MIL-STD-109	-	Quality Assurance Terms and Definitions
MIL-STD-810	-	Environmental Test Methods and Engineering Guidelines

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Military Specifications and Standards, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS (See 6.4)

US ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

10548052	-	Servo Drive, Traverse
10548062	-	Servo Drive, Elevation
10548193	-	Servo Electronics
10549134	-	Connector, Plug, Miniature
10556312	-	Antenna Assembly, AS-2049/VPS-2
10556313	-	Reflector and Feed Assembly

PACKAGING DATA SHEET

PSI0556312 - Packaging of Antenna Assembly, AS-2049/VPS-2

(Copies of drawings, publications and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

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2.2 Order of precedence. In the event of conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the applicable drawings and specifications.

3.2 Components and assemblies. The components and assemblies shall comply with all requirements specified on Drawing 10556312, all associated drawings, and with all requirements specified in applicable specifications and standards. Where applicable, the antenna assembly shall meet the requirements of MIL-F-13926 as follows:

a. Dimensions and tolerances.
b. Inorganic protective surface finishings.
c. Part identification and marking.
d. Workmanship.
e. Soldering electrical connections.

3.2.1 Subassemblies. The following subassemblies are covered by the indicated specifications:

Unit No.	Name	Specification
A1	Reflector and Feed Assembly 10556313	MIL-R-48427
A2	Servo Drive, Elevation: 10548062	MIL-S-50667
A3	Servo Drive, Traverse: 10548052	MIL-S-50666

3.3 Environmental service conditions.

3.3.1 Vibration. The antenna assembly shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after exposure to sinusoidal vibrations, in each of three mutually perpendicular axes, as follows:

Frequency Range	Amplitude
10 to 15 hertz	0.06 inches double amplitude
15 to 25 hertz	0.04 inches double amplitude
25 to 40 hertz	0.02 inches double amplitude

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The frequency of vibration shall be swept logarithmically.
The cycling time shall be 120 minutes per axis.
The sweep rate shall be 3.75 minutes ± 10% per decade.
The 10-40-10 hertz sweep time is 2 minutes 30 seconds ± 10%.

3.3.2 Shock. The antenna assembly shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after exposure to a total of 18 impulse shocks consisting of three shock impulses of 15 g's ± 10%, 11 milliseconds ± 10%, half-sine wave applied in each direction of 3 mutually perpendicular axes.

3.3.3 Altitude.

3.3.3.1 Operating. The antenna assembly shall meet the requirements of 3.4 and 3.5 during and after exposure to barometric pressure encountered at 10,000 feet above sea level (20.6 inches Hg).

Note: The requirements of 3.5.1, 3.5.2, 3.5.3 and 3.5.4 need not be checked during the exposure period.

3.3.3.2 Non-operating. The antenna assembly shall meet the requirements of 3.4 and 3.5 after exposure for not less than 10 minutes to barometric pressure encountered at 40,000 feet above sea level (5.5 inches Hg).

3.3.4 Temperature.

3.3.4.1 Operating. The antenna assembly shall meet the requirements of 3.4, 3.5, 3.5.5, 3.5.6 and 3.5.6.1 when stabilized thermally at any temperature between -40 deg. F and +155 deg. F.

3.3.4.2 Non-operating (storage). The antenna assembly shall not suffer permanent damage when subjected to the constant temperature extremes of -60 deg. F for a period of 24 hours and +160 deg. F for a period of 4 hours (after temperature stabilization), and shall meet the requirements of 3.4 and 3.5 when returned to room ambient.

3.3.5 Humidity (non-operating). The antenna assembly shall meet the requirements of 3.4 and 3.5 after exposure to an atmosphere with a relative humidity in excess of 95 percent for a period of 10 days.

3.4 Electrical. Each conductor shall be electrically continuous as specified on the schematic diagram of the antenna assembly.

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3.5 Performance. The antenna assembly shall meet the radio frequency characteristics when tested on an antenna test range which shall provide far field vision ($2D^2/\lambda$). This requires a minimum of 75 feet between transmitting and receiving antennas for the measurements. The antenna assembly shall be tested between the frequency range of 9.15 and 9.25 GHz.

3.5.1 Antenna pattern (without radome). The energy of any side lobe shall be at least 22 decibels (dB) below the main lobe energy.

3.5.1.1 Antenna pattern (with radome). The energy of any side lobe shall be at least 22 dB below the main lobe energy for the E-plane and not less than 21 dB for the H-plane.

3.5.2 Beamwidth. The beamwidth of the main lobe shall be 3.4 deg. to 4.3 deg. for energy received.

3.5.3 Gain. The antenna assembly shall have a gain of at least 31 dB without radome and at least 30 dB with radome over an isotropic radiator.

3.5.4 Boresight. The optical axis of the boresight clamp and the electrical center of the antenna beam shall be within $\pm$1/8 deg.

3.5.5 VSWR. The VSWR of the antenna assembly, when pointed in a nonreflecting direction, shall not be greater than 1.5:1.

3.5.6 Servo drive. The elevation and traverse servo drives shall be driven by servo electronics, 10548193, with synchro input signal which causes the antenna to oscillate sinusoidally $\pm$25 deg.

3.5.6.1 Servo drive dynamic test. When the elevation and traverse servos are driven at 0.1 Hz, the antenna shall track to less than 1.5 milliradians (mrad). The antenna performance shall be such that the "DEMOD" output signal (TP3) of the appropriate servo electronics, 10548193, when filtered to the 10 to 30 Hz passband and converted to a true root mean square (rms) signal by using item 17, and recorded as shown in Figure 1, shall not exceed 300 millivolts (mV). For the recorded waveforms shown in Figure 1, Channel 1 displays the filtered "DEMOD" output signal while Channel 2 displays the filtered and averaged signal of Channel 1 so as to provide a less complex waveform for quantitive analysis. Figure 1 shows (a) an acceptable antenna, (b) a nonacceptable antenna having a peak dc output exceeding 300 mV, and (c) a nonacceptable antenna having, in addition to greater than 300 mV level, visible vibrations present in the reflector and feed assembly (λ^2 E only).

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3.5.7 Function potentiometers.

3.5.7.1 Elevation. With +15 volts direct current (Vdc) applied to J1-g, J1-f, J1-E, J1-F, J1-M and J1-N of the elevation drive, the potentiometer outputs of λ_2 E at J1-S, λ_3 E at J1-P and E at J1-D shall provide the following voltages using J1-h as dc return:

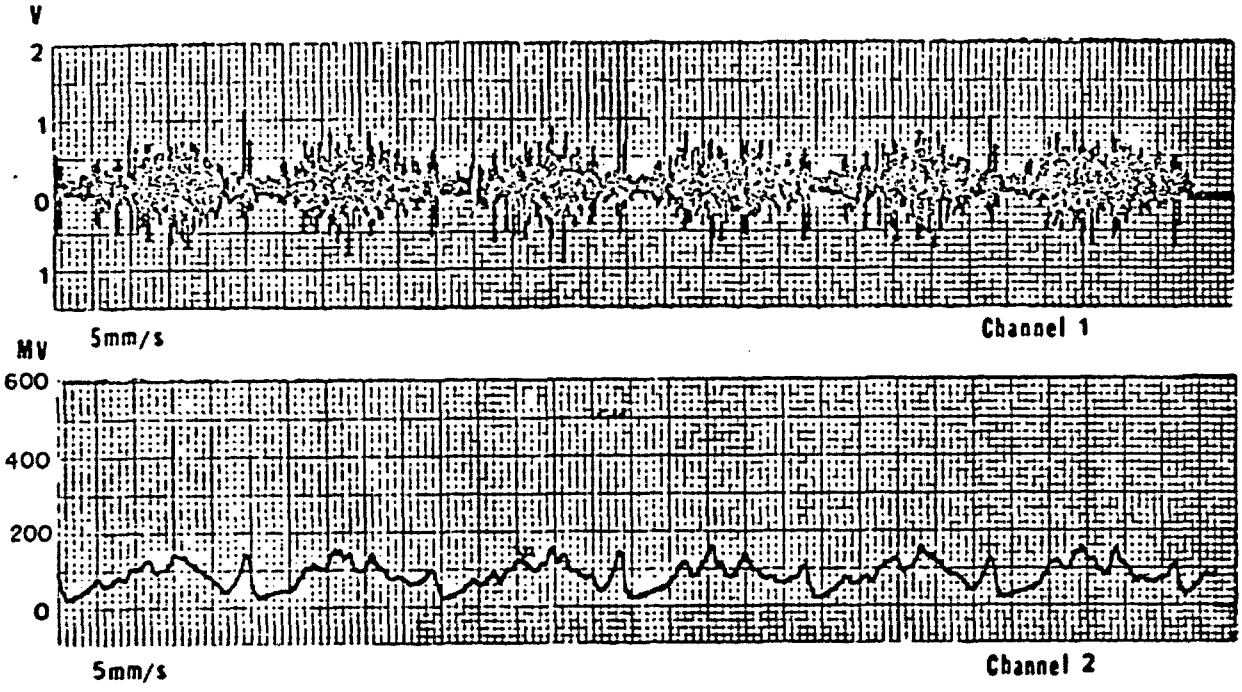
a. A null of not more than 2.50 volts at 0 deg. elevation lead angle (trunnion output) for the λ E potentiometer. A null of not more than 120 millivolts at 0 deg. elevation lead angle (trunnion output) for the λ_2 E and λ_3 E potentiometer output.

b. Measure the following voltage outputs at ±10 deg. elevation lead angle (corresponds to 12.25 deg. of synchro input information).

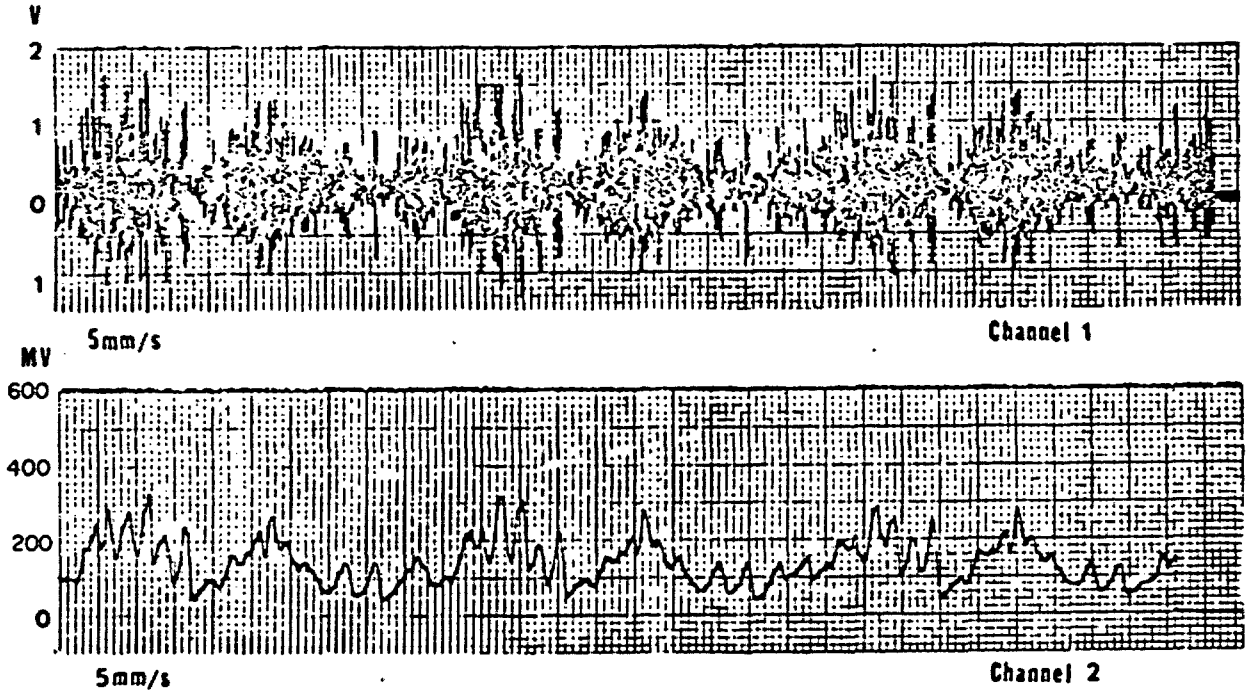
Signal	Voltage
λ E	7.40 volts maximum
λ_2 E	2.30 ± 0.4 volts
λ_3 E	0.90 ± 0.3 volts

c. The difference between the +10 deg. and -10 deg. lead angle voltages shall not exceed 0.2 V.

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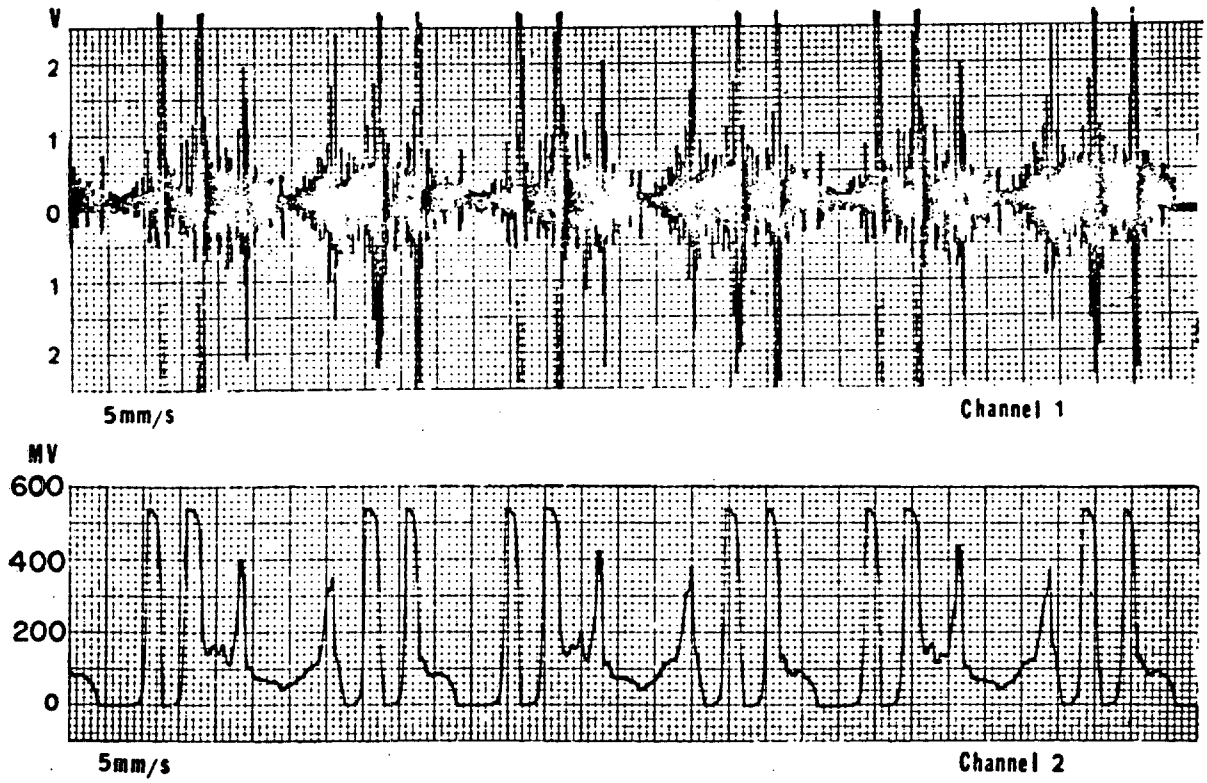
(a) Acceptable



(b) Non-acceptable

FIGURE 1. Chart recording displays (sheet 1 of 2).

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(c) Non-acceptable, visible vibiration

FIGURE 1. Chart recording displays (sheet 2 of 2).

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3.5.7.2 Traverse. With 15 Vdc applied to J1-g and J1-f of the traverse servo drive, the λ^2 T potentiometer output at J1-s shall provide the following voltages using J1-h as dc return:

a. A null of not more than 120 mV at 0 deg. traverse lead angle (trunnion output).

b. 2.4 ± 0.4 volts at ±10 deg. traverse lead angle (±10 deg. lead angle corresponds to ±12.7 deg. of synchro input information).

c. The difference between the 10 deg. and -10 deg. lead angle voltages shall not exceed 0.2 V.

3.6 First article inspection. When specified in the contract or purchase order, a sample shall be subjected to first article inspection in accordance with the technical provisions herein (see 4.3).

3.7 Workmanship. The workmanship requirements of MIL-S-45743, MIL-S-46844 and MIL-F-13926 shall apply.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define quality assurance terms used herein.

4.1.1 Responsibility for compliance. All items must meet all requirements of Sections 3 and 5 and all applicable drawings. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

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4.1.2 General provisions. The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

c. Special sampling inspection (see 4.5).

4.2.1 Inspection conditions. Unless otherwise specified, all examinations and tests shall be conducted at the following standard ambient conditions:

a.	Temperature	+50 deg. F to +90 deg. F
b.	Relative Humidity	90% RH Max.
c.	Pressure	28 to 32 inches Hg

4.3 First article inspection.

4.3.1 First article submission. The contractor shall submit a first article sample as designated by the contracting officer consisting of three (3) antenna assemblies for inspection in accordance with 4.3.2.

4.3.2 Inspections to be performed. First article assemblies, components and test specimens may be subjected by the Government to any or all of the examinations and tests specified in Table I and Table II herein and to any or all requirements of the applicable drawings.

4.3.3 Rejection. If any assembly, component or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon failure of any assembly, component or test specimen to comply with any of the stated requirements.

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. The term "inspection lot" is defined as a homogeneous collection of units of product from which a representative sample is drawn or which is inspected 100 percent to determine conformance with applicable requirements. Units of product selected for inspection shall represent only the inspection

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lot from which they are drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and the same specification revision. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with requirements.

4.4.2 Examination and tests.

4.4.2.1 Components, assemblies and subassemblies. All components, assemblies, and subassemblies shall be inspected in accordance with the inspection provisions contained in this specification and the Quality Assurance Provisions (QAP) listed in the technical data package (TDP). In the absence of QAPs, the applicable quality assurance provisions of MIL-F-13926 shall apply.

4.4.2.2 Final acceptance inspection. Subsequent to first article approval, examinations and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with the sampling plans specified in 4.4.2.3 herein. Examination and test for packaging and marking shall be in accordance with Section 5 herein. The tabulated classification of defects in 4.4.2.3 shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

4.4.2.3 Functional tests. The tests in Table I shall be performed on a 100% basis.

TABLE I. Functional tests (100 percent inspection).

	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURES
CRITICAL:		None	
MAJOR:			
101	Electrical	3.4	4.7.2
102	Performance	3.5	4.7.3
MINOR:		None	

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4.5 Special sampling. One antenna assembly out of the first 50 produced and one out of each 100 produced thereafter shall be selected at random. The sample shall have met the requirements and tests in Table I and shall then meet the requirements and tests in Table II. Unless otherwise specified in the contract, the items shall be returned to the lot upon successful completion of testing.

TABLE II. Special sampling.

CHARACTERISTIC		REQUIREMENT	TEST PROCEDURES
CRITICAL:		None	
MAJOR:			
103	Vibration	3.3.1	4.7.1.1
104	Shock	3.3.2	4.7.1.2
105	Altitude	3.3.3	4.7.1.3
106	Temperature	3.3.4	4.7.1.4
107	Humidity (non-operating)	3.3.5	4.7.1.5
MINOR:		None	

4.5.1 Failure of sample. Should any one item of a special sampling fail to meet any of the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections to all production since the last test and to current production has been accomplished in accordance with the provisions of the contract.

4.6 Inspection equipment.

4.6.1 Government-furnished inspection equipment. Where the contract provides for Government-furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements of MIL-I-45607.

4.6.2 Contractor-furnished inspection equipment.

4.6.2.1 Government design. Unless otherwise specified in the contract, all inspection equipment identified by drawing number in specifications or QAPs forming a part of the contract shall be supplied by the contractor in accordance with design specified. The contractor may, however, propose alternatives to Government designs for approval in accordance with 6.3

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4.6.2.2 Contractor design. The contractor shall design and supply inspection equipment compatible with the "Methods of Inspection" specified in 4.7 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926 wherever Government designs are not specified (see 4.6.2.1). Since tolerance of test equipment is normally considered to be within 10 percent of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions, and tolerances used in the design of test equipment shall be so selected and controlled as to ensure that the test equipment will reliably indicate acceptability of a product which does not exceed 90 percent of the prescribed limit and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of test equipment. Contractor inspection equipment designs shall be submitted for approval in accordance with 6.3.

4.6.2.3 Test equipment. In conjunction with 4.6.2.1 and 4.6.2.2, the standard test equipment listed in Table III shall be utilized in the performance of the applicable test methods of 4.7.

TABLE III. Test equipment list.

Item Number		Description
1.	Signal generator	Frequency: 9.15 to 9.25 GHz. RF power output: 1 and 100 mW. Modulator: AM modulation to 50% at 1 kHz, tunable ±10%.
2.	SWR meter and detector	Sensitivity: 0.15 V, tuned to 1 kHz;calibrated to read SWR with a square-law detector.
3.	Slotted line and detector	Frequency: 9.15 to 9.25 GHz. Residual SWR: Less than 1.05:1. Sensitivity: Sufficient to provide greater than 1 V at 1 kHz. Square law over greater than 20 dB.
4.	Attenuator	Attenuation: 10 dB SWR: 1 to 1.0 Frequency: 9.2 GHz
5.	Variable attenuator	Attenuation: 0 to 22 dB ±5%. Frequency: 9.2 GHz.

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TABLE III. Test equipment list (continued).

Item Number		Description
6.	Directional coupler	Coupling factor: 20 dB ± 1 dB. SWR: 1 to 1.0.
7.	Frequency meter	Frequency: 9.15 to 9.25 GHz. Accuracy: ±1%.
8.	Standard gain horn	Gain between 15 and 25 dB at 9.2 GHz.
9.	Antenna test range	Distance between antennas:75ft. Std. antenna gain: Sufficient to record VPS-2 antenna pattern, with minimum 30 dB dynamic range. Pedestal repeatability:±1 min. Pedestal azimuth travel:360 deg. Pedestal elevation travel:±20 deg. from horizon. Frequency: 9.15, 9.2 & 9.25 GHz. Recorder: Signal vs. direction ±1 min. Visual boresight: Target displaced in same direction from center of transmitting antenna by equal amount that boresight clamp is displaced from center of antenna assembly (11.25 ± .01 in).
10.	Test fixture (optional)	Fixture to mount antenna and return antenna in both planes, to within ±1 min of center between two electrical measurements.
11.	Servo control unit	See Figure 2. See Table IV for parts list.

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TABLE III. Test equipment list (continued).

Item Number		Description
12.	Dc power supply	24 Vdc, 10 A. Load regulation: ±1%. Ripple: Less than 100 mV.
13.	Dc power supply	15 Vdc, 100 mA. Load regulation: ±0.1%. Ripple: Less than 10 mV.
14.	Ac-dc voltmeter	Range: 500 mV to 150V. Input impedance: 10 megohms.
15.	Shaft clamp	See Figure 3.
16.	Servo director	Synchro transmitter compatible with 11CT4e. Simple harmonic motion of one cycle per 5 and 10 seconds with ±25 deg. adjustable sector scan.
17.	Rms voltmeter with dc output, Hewlett Packard 3400A or equivalent	Response: Responds to rms value (heatingvalue) of input signal. Frequency range: 10 Hz to 10 MHz. Fullscale accuracy: ±5% from 10 Hz to 50 Hz. Voltage range: 0.3 V to 3 V fullscale. Dc output: Ac to dc converter. Accuracy: ±5% for 10 Hz to 50 Hz. Output: 1 Vdc into open circuit for full scale deflection, proportional to meter deflection; 1 ma maximum. Nominal source impedance 1000 . Input impedance: 10M shunted by 50 PF. Output noise: 1 mV rms.

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TABLE III. Test equipment list (continued).

Item Number	Description
18. Band pass filter	Frequency response: 0.02 Hz to 20 kHz. Pass-band gain: 0 dB ± 1 dB. Attenuation slope: nominal 24 dB per octave. Maximum attenuation: greater than 80 dB. Input impedance: 22 megohms or greater. Parallel capacitance: 200 ± 50 pF.
19. Servo amplifier	In accordance with Drawing 10548193.
20. Recorder	Two channel. Sensitivity: 5mV/mm to 20 V/mm. Speed: 1.0 to 100 mm/s. Frequency response: 80 Hz.
21. Power source, 400 Hz	115 V, 1A with distortion less than 1%.

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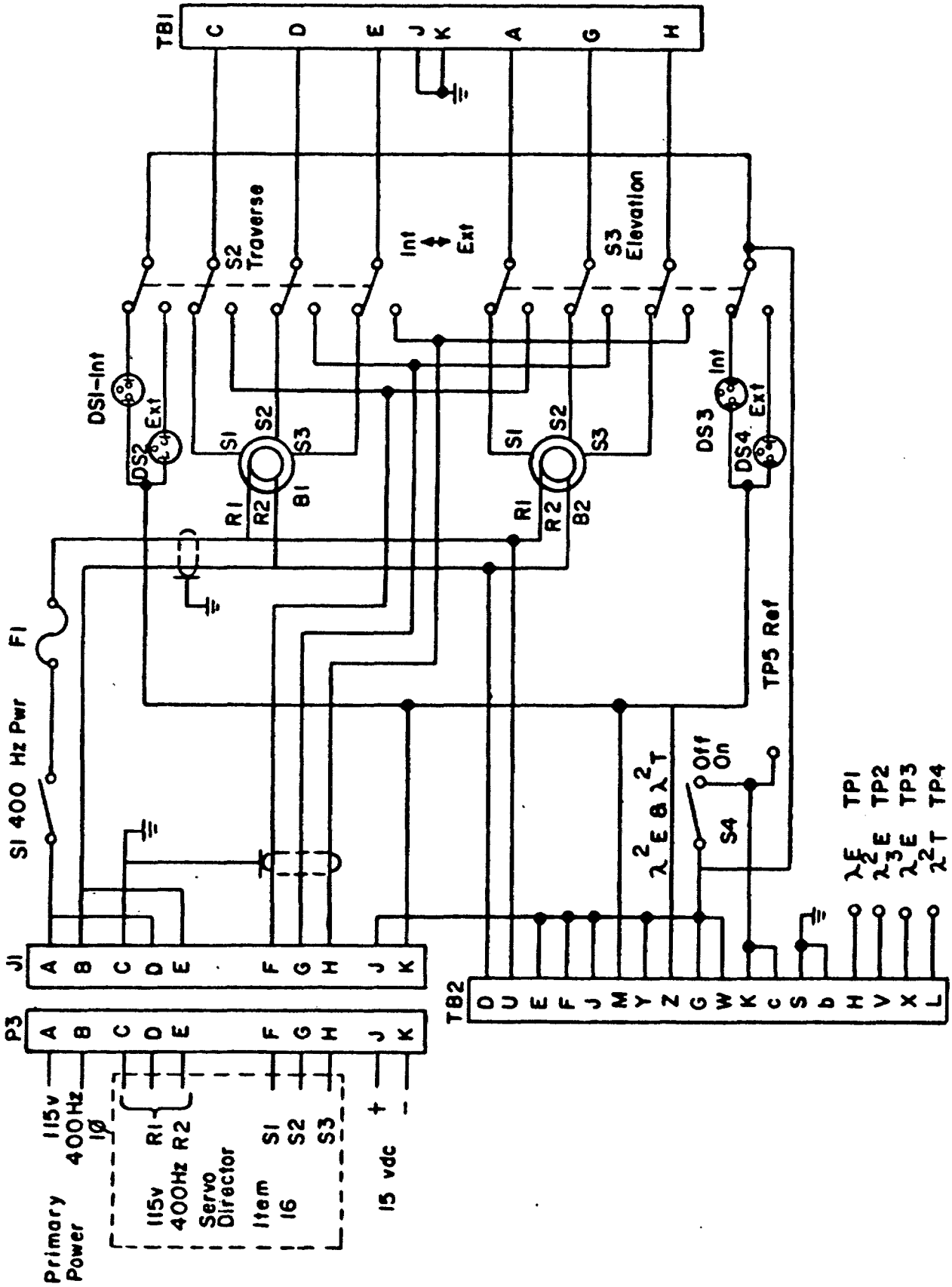


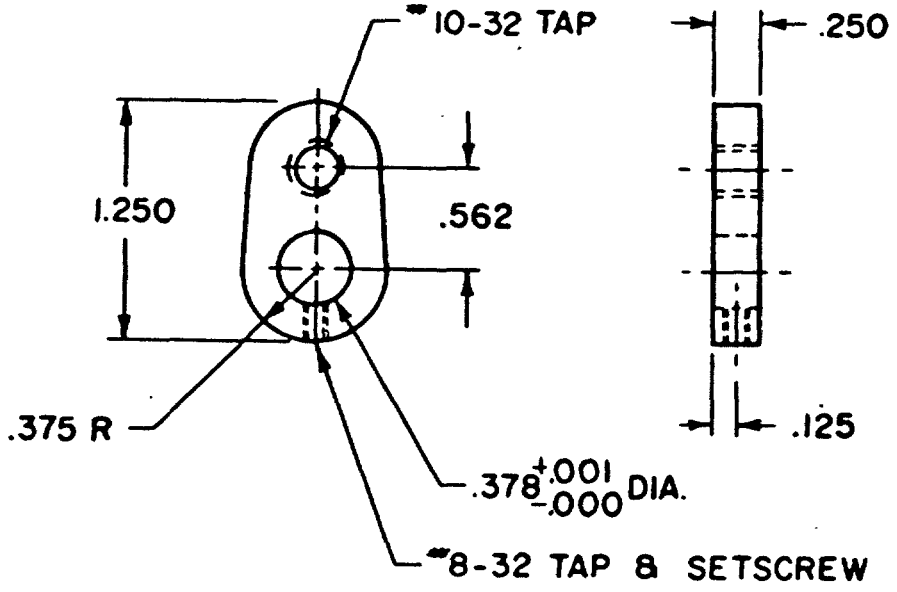
FIGURE 2. Servo Control Unit.

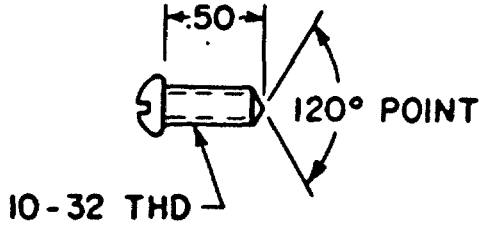
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TABLE IV. Servo control unit parts list.

Designation	Nomenclature and Description
J1	KPT02A-20-16P, chassis mounting
P1	KPT06B-16-26s, Cannon
TB1	Terminal board
TB2	Terminal board
B1	Synchro control transmitter, with dial
B2	Synchro control transmitter, with dial
S1, S4	Switch, SPST, 2A
S2, S3	Switch, 4PDT, 15A
F1, XF1	Fuse holder, and fuse, 1A
DS1 thru DS4	Neon glow lamp, NE-51
XDS1 thru XDS4	Lamp holder, Dialco 502-8136-102
TP1 thru TP5	Tip jack, H.H. Smith - 1501 or equivalent

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MATERIAL: CRES

FIGURE 3. Shaft clamp.

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4.7 Methods of inspection.

4.7.1 Environmental. Environmental tests shall be conducted in accordance with the test conditions, general test performance guidance, and individual test methods of MIL-STD-810, except as specified herein. Test facilities, chambers and apparatus used in conducting the tests contained in this specification shall be capable of meeting the conditions required.

4.7.1.1 Vibration. The test shall be performed as specified in Method 514.3, Category 8, Procedure I, of MIL-STD-810 as modified by 3.3.1 herein. In Step 1 of Procedure I only Steps 4 through 8 of paragraph II-2.1 apply. The antenna assembly shall be attached to the vibration generator directly or with a fixture, and securely held by its normal means of attachment. After vibration, the antenna assembly shall be tested as specified in 4.7.2 and 4.7.3 herein to determine compliance with 3.3.1.

4.7.1.2 Shock. This test shall be performed as specified in Procedure I, Section II of Method 516.3 of MIL-STD-810 except that:

a. The post test functional check (Step 5 of Procedure I) need not be conducted until completion of all shock impulses.

b. The shock pulses shall be 15 g's ± 10% amplitude and 11 ms ± 10% duration.

c. The maximum g shall occur at 5.5 milliseconds ±10%.

d. The shock measuring system shall have a flat frequency response within ±10% from 20 to 2,000 hertz.

After the application of all the shock impulses, the antenna assembly shall be tested as specified in 4.7.2 and 4.7.3 to determine compliance with 3.3.2.

4.7.1.3 Altitude (operating and non-operating). This test shall be performed as specified in Procedures II and III, Method 500.2 of MIL-STD-810. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.3.

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4.7.1.4 Temperature.

4.7.1.4.1 High (operating and non-operating). This test shall be performed as specified in Procedures I and II, Method 501.2 of MIL-STD-810 with constant temperature extremes as specified in 3.3.4. The operational checkout shall be as specified in 4.7.2, 4.7.3, for compliance with 3.3.4.

4.7.1.4.2 Low (operating and non-operating). This test shall be performed as specified in Procedures I and II, Method 502.2 of MIL-STD-810 with constant temperature extremes as specified in 3.3.4. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.4.

4.7.1.5 Humidity (non-operating). This test shall be performed in accordance with Procedure III, Method 507.2 of MIL-STD-810 except that in Step 7, checkout shall be performed at the conclusion of the test only, with the item at standard ambient conditions. The checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.5.

4.7.2 Electrical. Continuity shall be inspected by standard measuring equipment (see Table III) and visual examination for compliance with 3.4.

4.7.3 Performance. The selected antenna assemblies shall be tested as follows: The tests for gain, boresight, VSWR, servo drive, dynamic test, and function potentiometers (4.7.3.3, 4.7.3.4, 4.7.3.5, 4.7.3.6 and 4.7.3.7) shall be performed on all antenna assemblies produced. The tests for antenna pattern (4.7.3.1) and beamwidth (4.7.3.2) need only be performed on the first article and at least 10 percent of production equipment. The tests for gain, boresight, antenna pattern and beamwidth (4.7.3.3, 4.7.3.4, 4.7.3.1 and 4.7.3.2) may be performed on the reflector and feed subassembly (Drawing 10556313) per requirement, at the discretion of the contractor.

4.7.3.1 Antenna pattern. The antenna pattern is to be obtained as follows:

a. Connect signal generator, item 1, to antenna test range transmitting antenna (part of item 9) through variable attenuator, item 5, and adjust frequency for 9.2 gigahertz (GHz).

b. Use test fixture, item 10, to mount antenna assembly to pedestal and receiver of antenna test range (part of item 9) and point antenna assembly toward test transmitting antenna. Adjust antenna test range

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equipment controls for a peak on test range recorder while turning, in all directions, antenna assembly under test for maximum recorder deflection.

c. Adjust test range equipment for full recorder deflection.

d. Rotate antenna under test to 180 deg. (in azimuth) from transmitting antenna and make a 360 deg. azimuth recording of antenna assembly under test. Return antenna to peak deflection position of antenna test range and increase variable attenuator setting by 22 dB.

e. Note the recorder deflection. Decrease variable attenuator setting by 22 dB. Rotate the antenna to all side lobes. Note maximum recorder deflection for each side lobe. The side lobes of the antenna assembly as recorded above shall not be greater than the 22 dB down reference level.

f. Repeat a. to obtain full deflection. Rotate antenna under test in a vertical direction to 20 deg. from horizontal. Record pattern in elevation ±20 deg. from horizon. Return antenna to peak deflection of recorder and increase variable attenuator attenuation by 22 dB.

g. Note the recorder deflection. Decrease variable attenuator setting by 22 dB. Rotate the antenna vertically to all side lobes. Note maximum recorder deflection for each side lobe. The side lobes of the antenna assembly as recorded above shall not be greater than the 22 dB down reference level.

h. Repeat a. through g. for frequencies 9.15 and 9.25 GHz.

4.7.3.2 Beamwidth. The beamwidths will be obtained by repeating 4.7.3.1a for 9.2 GHz.

a. Operate recorder to obtain expanded directional pattern (i.e., 6 deg. to 60 deg.).

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b. Start azimuth pattern at least 3 deg. deflection from peak obtained above and record at least 3 deg. past peak.

c. Return antenna assembly to peak and adjust variable attenuator to obtain 3 dB attenuation.

d. Use this level to mark antenna pattern with 3 dB reference.

e. The 3 dB beamwidth as obtained shall be between 3.4 deg. and 4.3 deg.

f. Repeat a. through e. except rotation will be in elevation.

g. The 3 dB beamwidth as obtained shall be between 3.4 deg. and 4.3 deg.

h. Repeat a. through h. for frequencies of 9.15 and 9.25 GHz.

i. The 3 dB beamwidth as obtained shall be between 3.4 deg. and 4.3 deg.

4.7.3.3 Gain. The gain shall be obtained by performing 4.7.3.1a for 9.2 GHz using antenna test range or optional test fixture (no recording required).

a. Adjust the antenna test range equipment to obtain a variable attenuation of at least 3 dB greater than 31 dB, minus the "standard gain horn" gain when full recorder deflection or similar indication is obtained.

b. The antenna assembly under test is replaced with standard gain antenna and variable attenuator adjusted to obtain full recorder deflection.

c. The change in variable attenuator setting plus the gain of the standard gain antenna shall be no less than 31 dB.

4.7.3.4 Boresight. The boresight of the antenna shall be accomplished using the antenna test range (item 9) or holding fixture which allows movement of the antenna, plus/minus 4 1/2 minutes from center and return to center within ±1 minute from electrical measurement.

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a. The antenna shall be adjusted for electrical center (peak) by repeating 4.7.3.1a for 9.2 GHz using antenna test range or test fixture.

b. Adjust variable attenuator to increase attenuation by 3 dB and record level of recorder deflection.

c. Return variable attenuator to value before 3 dB increase.

d. Rotate antenna assembly until recorded level of 3 dB above is passed in azimuth.

e. Return antenna assembly back to obtain 3 dB reference above (above operations to overcome backlash).

f. Note reading or indication and continue rotation through peak and to next 3 dB reference. Note reading.

g. Move antenna assembly to halfway between the two 3 dB readings taking care to go past desired reading and return so approach is from same direction as before.

h. Repeat d. for elevation. Place telescope in boresight telescope holder and observe boresight target in cross hairs.

4.7.3.5 VSWR. Standing wave measurements are obtained by connecting signal generator, item 1, output to slotted line and detector, item 3, with attenuator, item 4, between item 1 and 3.

a. Connect SWR meter, item 2, to detector output.

b. Connect output of slotted line to antenna assembly under test and aim antenna in a nonreflecting direction.

c. Turn equipment on and allow to stabilize.

d. Adjust frequency of signal generator to 9.2 GHz.

e. Adjust the modulation frequency of signal generator to obtain a peak reading of the SWR meter.

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f. Adjust the signal generator attenuator level and slotted line probe for a full scale reading on the SWR meter and so that all measurements are made within the square law region of the slotted line detector.

g. Obtain the SWR as indicated on the SWR meter by movement of the slotted line probe for maximum to minimum voltage levels.

h. Repeat e. for 9.15 and 9.25 GHz. SWR shall be less than 1.5 to 1.

4.7.3.6 Servo drive dynamic test. The servo drives shall be tested as follows:

a. Connect the antenna assembly to the test equipment as shown in Figure 4. Connect the servo electronics, item 19, to W1 and filter as shown by the solid outline in Figure 4, for the performance test on the elevation servo drive of the antenna.

b. The antenna assembly shall be mounted to a rigid platform during the following tests. Install the shaft clamp, item 15, to the elevation drive shaft and with the servo control unit, item 11, in the internal mode of operation set the elevation dial to 0 deg. Apply power to the antenna and adjust the shaft so the antenna is approximately 25 deg. above the horizontal, then lock in place with the shaft clamp.

c. Set the servo control unit to the external mode and adjust the servo director so that it is oscillating 25 deg. either side of 0 deg. at a 0.1 Hz rate.

d. Set the recorder, item 20, to operate at a chart speed of 5 millimeters per second (5 mm/s); Channel 1 gain at 1 volt per centimeter (1 V/cm) and Channel 2 gain at 20 millivolts per millimeter (20 mV/mm). Establish a zero reference for Channel 1 at the center of the chart and a zero reference for Channel 2 near the edge of the chart (see Figure 1).

e. Set the filter, item 18, for a band pass of 10 Hz to 30 Hz.

f. Set the true rms voltmeter, item 17, to the 1-volt scale.

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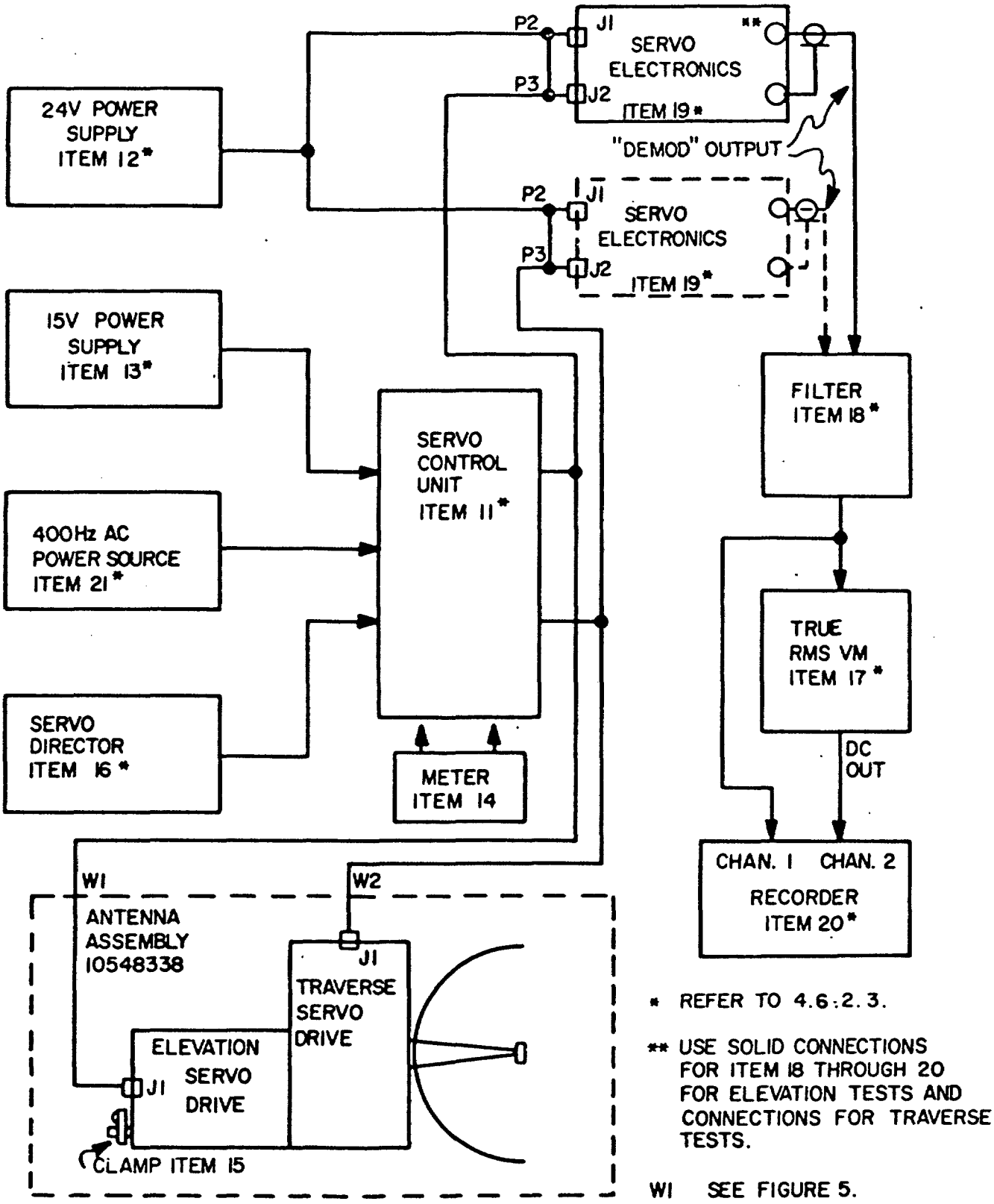
g. With the antenna scanning ±25 deg. of 0 deg. reference record four cycles of antenna operation. The maximum value of the signal as recorded on Channel 2 (rms voltmeter dc output) shall be 300 mV.

NOTE: In addition to the 300 mV limit being exceeded, the basic criteria for rejection of an antenna servo drive is chatter or vibration of the antenna dish or feed horn readily discernible by observation. Figure 1(c) displays a chart recording of such a condition.

h. For the performance test on the traverse servo drive, connect the servo electronics, item 19, to W2 and perform steps c. through g.

4.7.3.7 Function potentiometers. For test of the function potentiometers set up antenna assembly as shown in Figure 4. Synchros B1 of each servo drive shall have been adjusted for electrical 0 deg. Operate the servo control unit, item 11, to the internal mode of operation, and turn S4 to the on position.

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* REFER TO 4.6.2.3.

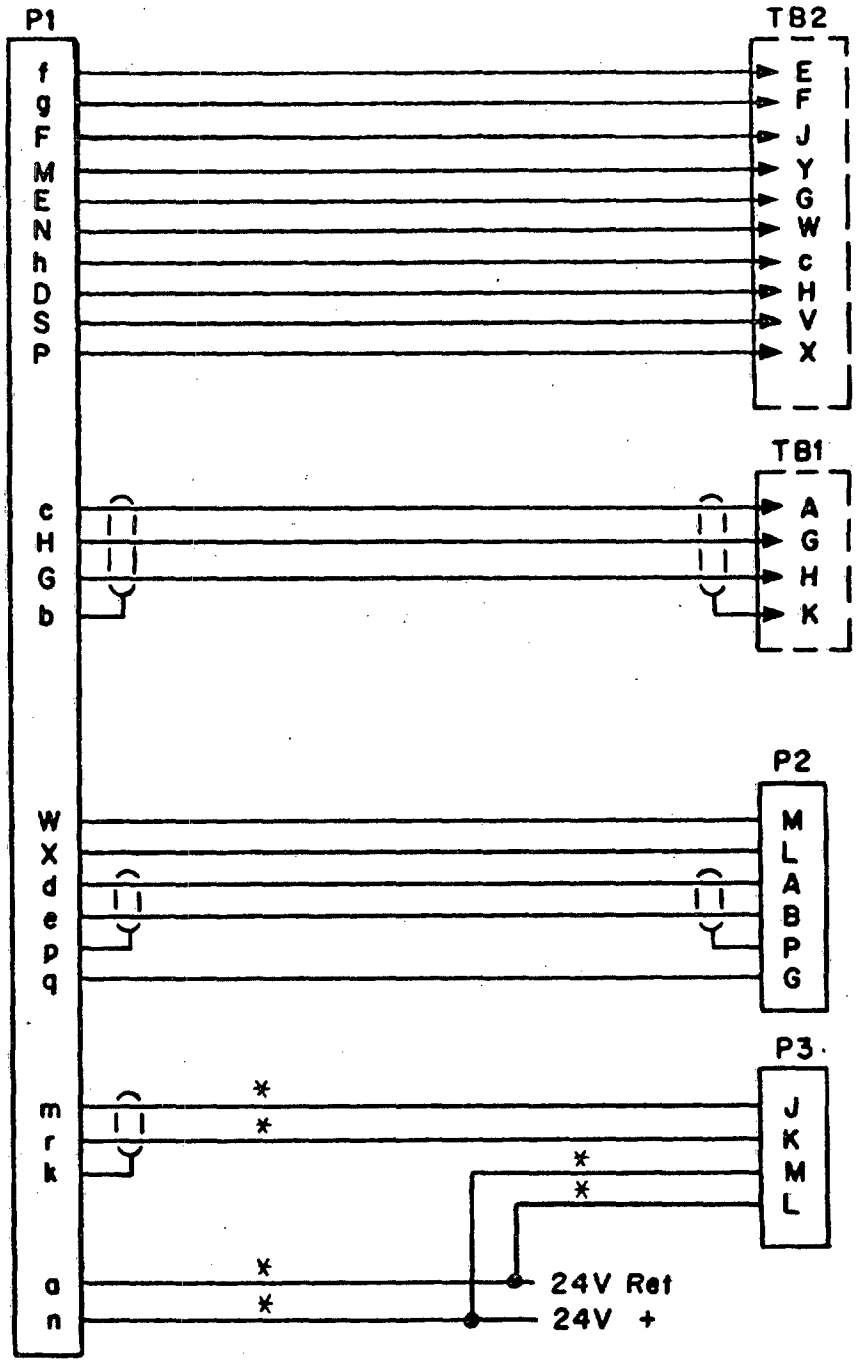
** USE SOLID CONNECTIONS FOR ITEM 18 THROUGH 20 FOR ELEVATION TESTS AND CONNECTIONS FOR TRAVERSE TESTS.

W1 SEE FIGURE 5.

W2 SEE FIGURE 6.

FIGURE 4. Test setup.

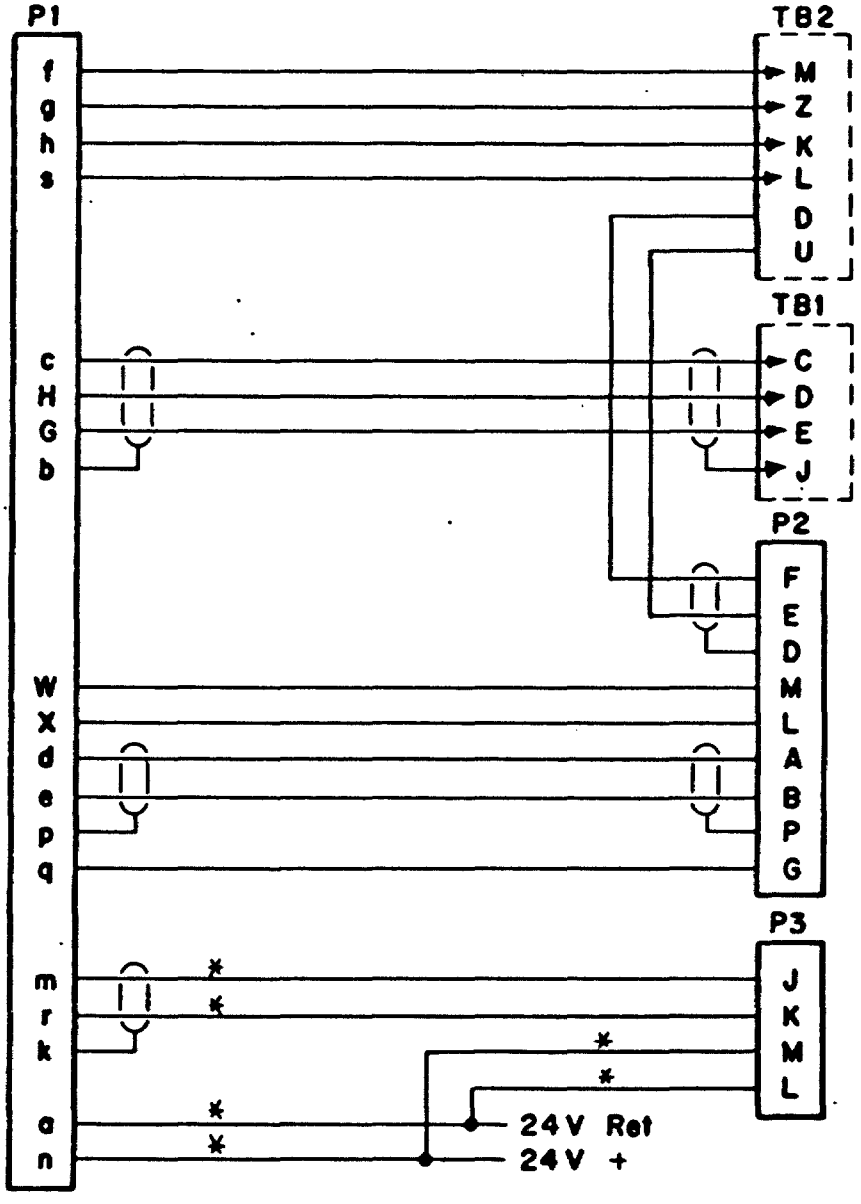
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* CONDUCTOR SIZE 16 AWG, ALL OTHERS 22 AWG.
P1 - CONNECTOR, PLUG ARDEC PART NUMBER 10549134-8
P2 - CONNECTOR, PLUG - MS3116E-14-19-S
P3 - CONNECTOR, PLUG - MS3116C-14-12-S
TB1 & TB2 - PART OF SERVO CONTROL UNIT 11.

FIGURE 5. Test cable W1.

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* CONDUCTOR SIZE 16 AWG, ALL OTHERS 22 AWG.
P1- CONNECTOR, PLUG ARDEC PART NUMBER 10549134-7
P2- CONNECTOR, PLUG-MS3116E-14-19-S
P3- CONNECTOR, PLUG-MS3116E-14-12-S
TB1 & TB2 PART OF SERVO CONTROL UNIT ITEM 11.

FIGURE 6. Test cable W2.

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4.7.3.7.1 Elevation. Using the voltmeter, item 14, at TP1, TP2 and TP3 on the servo control unit, measurement of the λ E, λ^2 E and λ^3 E potentiometer outputs is as specified in 3.5.7.1. Use TP5 on the servo control unit as reference for voltage measurements. When the lead angle is specified as ±10 deg., position the dial of the servo control unit to ±12.25 deg.

4.7.3.7.2 Traverse. Using the voltmeter at TP4 on the servo control unit, measure λ^2 T potentiometer output to be as specified in 3.5.7.2. Use TP5 on the servo control unit as reference for voltage measurements. When the lead angle is specified as ±10 deg., position the dial of the servo control unit to ±12.7 deg.

5. PACKAGING

5.1 General. Preservation, packing and marking shall be in accordance with Packaging Data Sheet PS10556312. The level of protection shall be as specified in the procurement document.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This specification details the specific requirements for design, performance, quality assurance and testing for the antenna assembly of the AN/VPS-2 Radar Set.

6.2 Acquisition requirements . Procurement documents shall specify the following.

6.2.1 Procurement requirements.

a. Title, number and date of this specification.

b. Applicable stock number.

c. First article sample requirements.

d. Packaging requirements, if other than specified in Section 5.

e. Serialization requirements, if applicable.

f. Certificate of Conformance for each lot or shipment of product.

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6.3 Submission of contractor inspection equipment designs for approval. Submit two copies of designs as required to: Commander, U.S. Army AMCCOM, Attn: AMSMC-QAF-I, Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract. Unless otherwise specified, data item DI-R-1714 will apply.

6.4 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament Research Development and Engineering Center (ARDEC), may also include drawings prepared by, and identified as, U.S. Army Armament Research and Development Command (ARRADCOM), Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under cognizance of ARDEC.

6.5 Subject term (Key word) listing.

Radar Set AN/VPS-2
Servo electronics
Reflector and feed
Antenna assembly
VSWR
Function potentiometers
Servo control

6.6 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1285-A663)

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